4.3.51

EE25BTECH11061 - V.Sainadh

Question: The equation of a line is $3x - 4y + 10 = 0$. Find its slope and the x and y intercepts.

Solution: Let $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$. Then

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} = -10 \tag{1}$$

Comparing with $\mathbf{n}^T\mathbf{x} = c$, we identify

$$\mathbf{n} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}, \qquad c = -10. \tag{2}$$

A direction vector orthogonal to $\mathbf{n}$ is

$$\mathbf{m} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \quad (\mathbf{n}^T\mathbf{m} = 3 \cdot 4 + (-4) \cdot 3 = 0). \tag{3}$$

Hence the slope is

$$\text{slope} = \frac{m_y}{m_x} = \frac{3}{4}. \tag{4}$$

x-intercept: points on the x-axis have $\mathbf{x} = \begin{pmatrix} x \\ 0 \end{pmatrix}$. Substituting in (1),

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \begin{pmatrix} x \\ 0 \end{pmatrix} = -10 \implies 3x = -10 \implies x = -\frac{10}{3}. \tag{5}$$

$$\text{x-intercept} = \begin{pmatrix} -\frac{10}{3} \\ 0 \end{pmatrix}. \tag{6}$$

y-intercept: points on the y-axis have $\mathbf{x} = \begin{pmatrix} 0 \\ y \end{pmatrix}$. Substituting in (1),

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \begin{pmatrix} 0 \\ y \end{pmatrix} = -10 \implies -4y = -10 \implies y = \frac{5}{2}. \tag{7}$$

$$\text{y-intercept} = \begin{pmatrix} 0 \\ \frac{5}{2} \end{pmatrix}. \tag{8}$$

Graph:

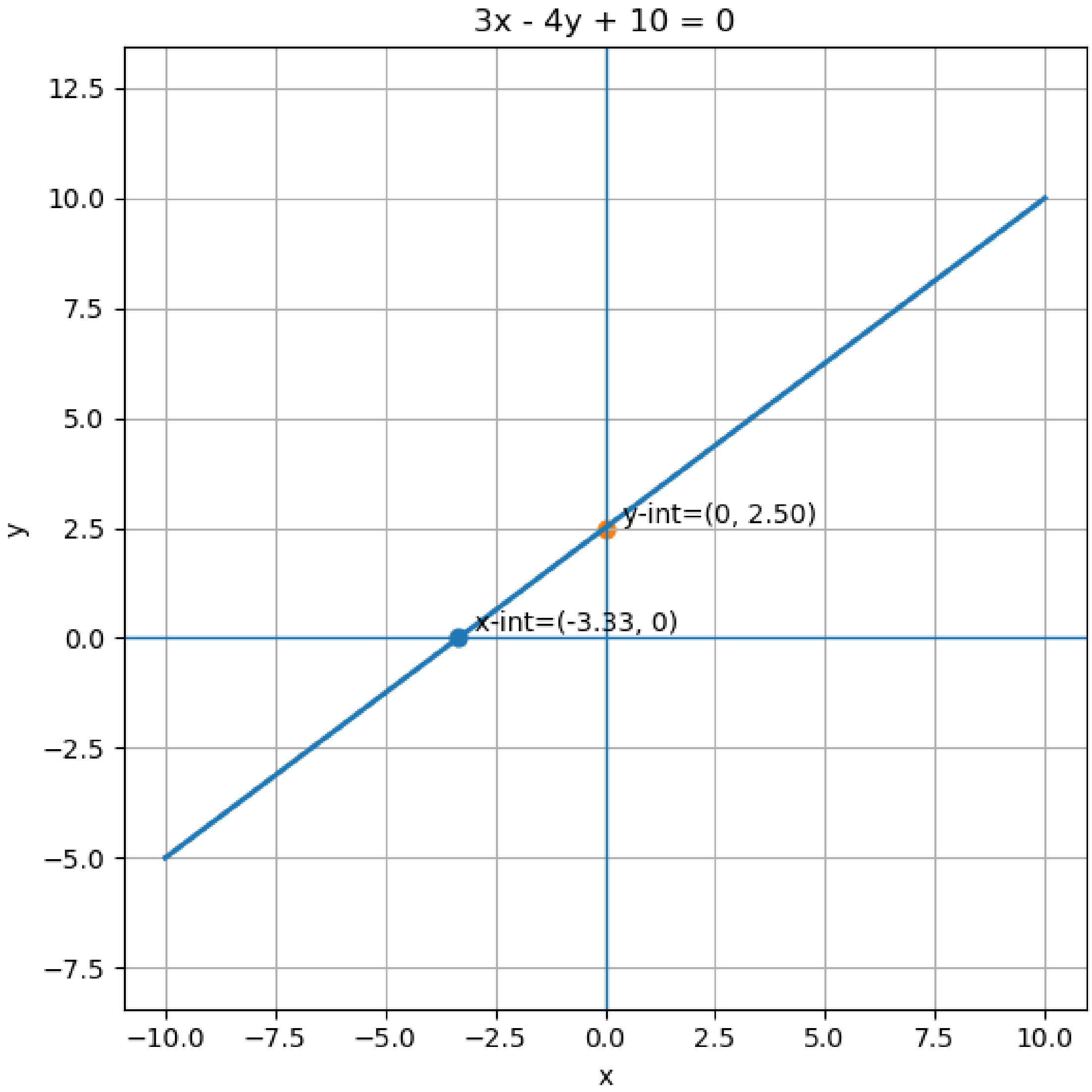

Fig. 0: Plot of the line $3x - 4y + 10 = 0$